Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	22 February 2026
Team ID	LTVIP2026TMIDS42033
Project Name	**Weather-Based Prediction of Wind Turbine Energy Output: A Next-Generation Approach to Renewable Energy Management**
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Weather API Integration	USN-1	As a system, I can fetch real-time weather data using OpenWeatherMap API	3	High	Team
Sprint-1	Weather API Integration	USN-2	As a developer, I can configure API key securely	2	High	Team
Sprint-1	ML Model Development	USN-3	As a system, I can preprocess weather data for training	3	High	Team
Sprint-1	ML Model Development	USN-4	As a system, I can train regression model to predict energy output	5	High	Team
Sprint-2	Backend Integration	USN-5	As a user, I can request energy prediction through REST API	3	High	Team
Sprint-2	Frontend Dashboard	USN-6	As a user, I can view real-time weather data in dashboard	2	Medium	Team
Sprint-2	Frontend Dashboard	USN-7	As a user, I can see predicted wind energy output	3	High	Team
Sprint-2	Analytics	USN-8	As a user, I can view energy charts & gauge visualization	5	Medium	Team

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	**16**	7 Days	01 Feb 2026	07 Feb 2026	16	07 Feb 2026
Sprint-2	**18**	7 Days	10 Feb 2026	16 Feb 2026	18	16 Feb 2026

Velocity Calculation

Total Story Points = 34

Number of Sprints = 2

Velocity = 34 / 2

= 17 Story Points per Sprint

Burndown Chart Explanation

A burndown chart represents remaining work versus time during sprint execution. It helps monitor development progress and ensures timely delivery of the Wind Energy Prediction System.

[1] Visual Paradigm, “Scrum Burndown Chart,” *Visual Paradigm Scrum Guide*. [Online]. Available: https://www.visual-paradigm.com/scrum/scrum-burndown-chart/.

[2] Atlassian, “Burndown charts,” *Atlassian Agile Tutorials*. [Online]. Available: https://www.atlassian.com/agile/tutorials/burndown-charts.

Reference:

[1] Atlassian, “Agile project management,” *Atlassian Agile Guides*. [Online]. Available: https://www.atlassian.com/agile/project-management.

[2] Atlassian, “How to do Scrum with Jira Software,” *Atlassian Agile Tutorials*. [Online]. Available: https://www.atlassian.com/agile/tutorials/how-to-do-scrum-with-jira-software.

[3] Atlassian, “Epics,” *Atlassian Agile Tutorials*. [Online]. Available: https://www.atlassian.com/agile/tutorials/epics.

[4] Atlassian, “Sprints,” *Atlassian Agile Tutorials*. [Online]. Available: https://www.atlassian.com/agile/tutorials/sprints.

[5] Atlassian, “Estimation,” *Atlassian Agile Project Management*. [Online]. Available: https://www.atlassian.com/agile/project-management/estimation.

[6] Atlassian, “Burndown charts,” *Atlassian Agile Tutorials*. [Online]. Available: https://www.atlassian.com/agile/tutorials/burndown-charts.